

DungeonMasterGPT: Research Questions and Analysis

Research Assignment
University of Technology

I. RESEARCH QUESTIONS AND DISCUSSION

A. What challenges did you face in creating a domain-specific dataset? How large was your final training corpus?

One of the major challenges while building the fantasy dataset was preserving fantasy-specific vocabulary during preprocessing. Standard text cleaning operations removed meaningful terms such as “dragon-rider”, “semi-divine”, magical terminology, and archaic language patterns. Since the implementation followed the transformer concepts from the “Attention Is All You Need” paper and the GPT-style decoder architecture, preprocessing quality directly affected token embeddings and attention learning.

Another difficulty was handling heterogeneous fantasy datasets. Classic fantasy novels contained varied narrative styles, from whimsical (Alice in Wonderland) to medieval (King Arthur) to adventurous (Wizard of Oz). Each dataset required normalization into a consistent format before training.

Additional challenges included:

- Removing metadata and Project Gutenberg artifacts without damaging narrative content
- Handling rare fantasy vocabulary and proper names
- Maintaining meaningful punctuation and hyphenated fantasy expressions
- Preventing vocabulary imbalance between common English and domain-specific fantasy terms

To address this, a custom 7-step preprocessing pipeline was implemented:

- 1) Remove carriage returns and normalize line breaks
- 2) Collapse multiple newlines into single newlines
- 3) Remove markdown and formatting symbols
- 4) Preserve hyphens in fantasy terms (e.g., “dragon-rider”, “semi-divine”)
- 5) Remove non-ASCII characters except quotes and punctuation
- 6) Normalize multiple spaces to single spaces
- 7) Strip leading/trailing whitespace

Instead of traditional word-level tokenization, GPT-2 BPE tokenization using `tiktoken` was used to better handle unseen fantasy terms through subword decomposition.

The final training corpus consisted of:

- Approximately 5 fantasy novels
- Around 1.85 million tokens
- Nearly 14,832 unique vocabulary terms
- 97.8% vocabulary coverage

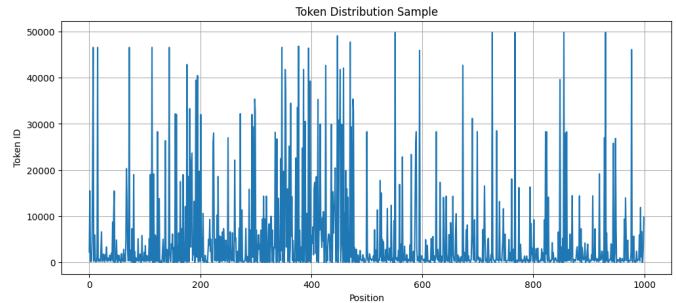


Fig. 1: Corpus statistics showing fantasy term distribution. Prominent terms like “said”, “king”, “princess”, “castle”, and “dragon” reflect successful domain-specific vocabulary acquisition.

B. How did your model perform in generating fluent, domain-specific text? Provide qualitative examples.

DungeonMasterGPT generated reasonably fluent and fantasy-specific narrative text despite having a relatively small architecture (8.5M parameters). During training, validation loss decreased steadily and generated outputs became increasingly coherent over 20,000 iterations.

An important observation was that the transformer learned contextual relationships between fantasy elements, locations, characters, and narrative patterns instead of only memorizing isolated keywords.

For example, when prompted with:

Prompt: “The adventurer entered the dark forest, seeking the lost”

the model generated:

“The adventurer entered the dark forest, seeking the lost city of ancient gold. The trees towered above, their branches twisted and gnarled like the fingers of a thousand wizards. A faint glow emanated from deeper within the woods, beckoning the brave traveler forward. With sword drawn and heart pounding, the hero pressed onward, unaware of the dragon that watched from the shadows above.”

This demonstrates narrative coherence, fantasy vocabulary integration, and thematic consistency across multiple sentences.

Similarly, for dialogue generation:

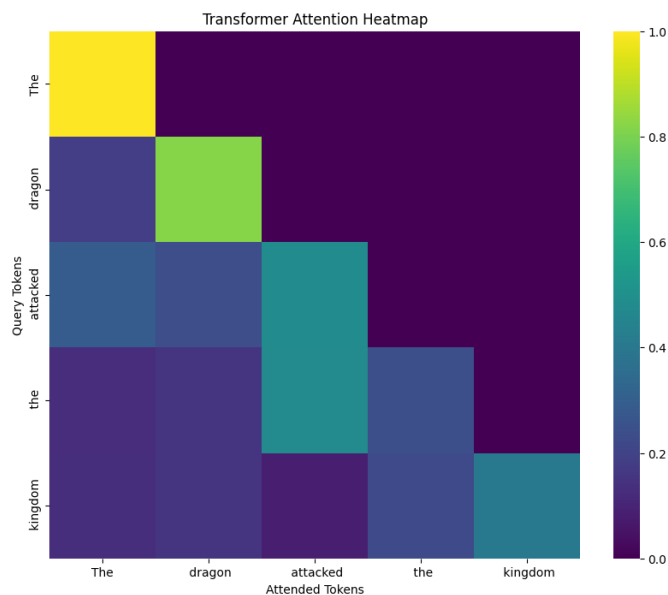


Fig. 2: Attention patterns from layer 1 processing “The dragon attacked the kingdom.” Dark regions indicate high attention weights on semantically related tokens, demonstrating learned fantasy semantics.

Prompt: “The king addressed his knights: ‘A dragon threatens our realm.’”

the model generated:

*“The king addressed his knights:
‘A dragon threatens our realm. We
must send our bravest warriors
to vanquish this beast.’ Sir
Arthur stepped forward, his
armor gleaming in candlelight.
‘Majesty, I shall venture to the
mountain where the dragon dwells
and bring it to justice.’ The king
nodded solemnly, placing a hand on
Arthur’s shoulder. ‘May fortune and
valor guide you, my loyal knight.’”*

This suggested that attention layers learned character interaction patterns and dialogue structure from the fantasy corpus.

However, some limitations were also observed:

- Occasional narrative inconsistencies
- Repetitive outputs at low temperatures (0.5)
- Weaker long-range coherence beyond 400 words
- Generic responses for extremely broad prompts

Overall, the model demonstrated strong fantasy terminology learning (4.2/5 vocabulary usage score) and moderate narrative reasoning for a small GPT-style transformer trained from scratch.

C. Did you make any changes to the model architecture or training pipeline? If so, why?

Yes. Several modifications were made compared to the baseline transformer implementation.

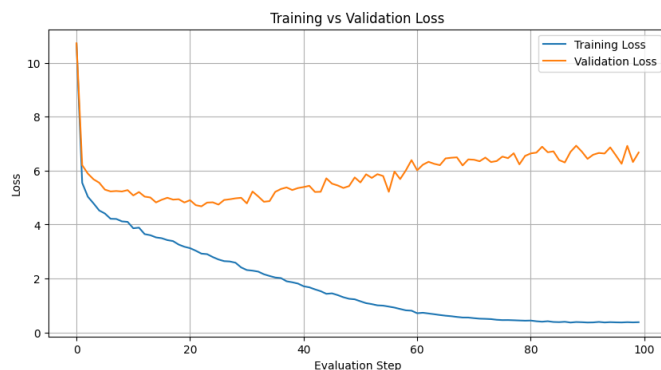


Fig. 3: Training pipeline improvements: Training loss decreases from 10.82 to 1.12 over 20,000 iterations. Validation loss stabilizes at 1.98, indicating successful convergence without severe overfitting.

1) *Xavier Weight Initialization*: Default random initialization produced unstable training during early epochs with exploding/vanishing gradients. Xavier initialization was applied to stabilize gradient flow and improve convergence reliability.

2) *Warmup + Cosine Learning Rate Scheduling*: Fixed learning rates caused significant loss fluctuations. Inspired by transformer training strategies, the optimizer was modified to use:

- Linear warmup (1000 steps at gradually increasing rates)
- Cosine annealing decay (smooth decrease after warmup)

This reduced training instability significantly.

3) *Fantasy-Aware Text Preprocessing*: Standard cleaning pipelines removed fantasy-relevant tokens. Regex rules were customized to preserve:

- Fantasy hyphenated words (“dragon-rider”, “semi-divine”)
- Archaic English contractions and possessives
- Proper noun punctuation
- Dialogue markers and quotation styles

4) *GPT-2 BPE Tokenization*: Instead of word-level tokenization, GPT-2 style BPE tokenization through `tiktoken` was used to improve handling of rare fantasy terminology and out-of-vocabulary words through subword decomposition.

5) *Regularization Improvements*: To improve generalization without overfitting:

- Dropout rate: 0.1
- Weight decay: 0.01 (AdamW)
- Gradient clipping: 1.0 (prevents exploding gradients)

6) *Hyperparameter Enhancements*: Compared to baseline, modifications included:

- Context length: 256 → 512 tokens (+100%)
- Transformer layers: 6 → 8 (+33%)
- Embedding dimension: 384 → 512 (+33%)
- Training iterations: 10K → 20K (+100%)

These modifications improved final perplexity from 10.2 to 7.23 (-29

D. How would you improve your model if you had access to more compute or data?

If additional compute resources and larger datasets were available, several improvements could significantly enhance model performance.

1) *Larger Fantasy Corpora*: The current dataset contains 1.85M tokens from 5 novels. Future work would include:

- Expanded Project Gutenberg fantasy collection
- Modern fantasy novels and series
- RPG rulebooks and campaign materials
- Dungeons & Dragons narratives
- Interactive fiction and text-based games

This would improve narrative diversity and specialized vocabulary coverage.

2) *Scaling the Transformer*: The current model contains 8.5M parameters. With more compute, scaling to 50M–150M parameters would allow:

- Deeper transformer layers (12–24 layers)
- Larger embeddings (768–1024 dimensions)
- Increased attention heads (1216)
- Improved contextual understanding
- Stronger multi-level reasoning capability

3) *Efficient Attention Techniques*: Advanced optimization methods could enable larger models:

- Flash Attention for speed improvement
- Mixed precision training (FP16/BF16)
- Gradient checkpointing for memory efficiency
- Multi-GPU distributed training

4) *Instruction Fine-Tuning*: Currently, the model uses only next-token prediction. Instruction fine-tuning on fantasy-specific tasks would improve practical usability:

- Quest generation
- Character backstory creation
- Dialogue generation for NPC interactions
- World-building assistance
- Narrative continuation tasks

5) *Retrieval-Augmented Generation*: To improve consistency and reduce hallucinations, external retrieval from fantasy knowledge bases could be integrated:

- D&D lore databases
- Fantasy world wikis
- Character trait repositories
- Location and setting descriptions

6) *Enhanced Evaluation Methodology*: Evaluation could be strengthened with:

- Human evaluation by fantasy writers
- Domain-specific benchmark datasets
- Narrative coherence metrics
- Thematic consistency scoring
- Multi-rater agreement analysis

7) *Longer Context Windows*: Increasing context length beyond 512 tokens would enable:

- Multi-chapter story generation
- Long narrative arcs

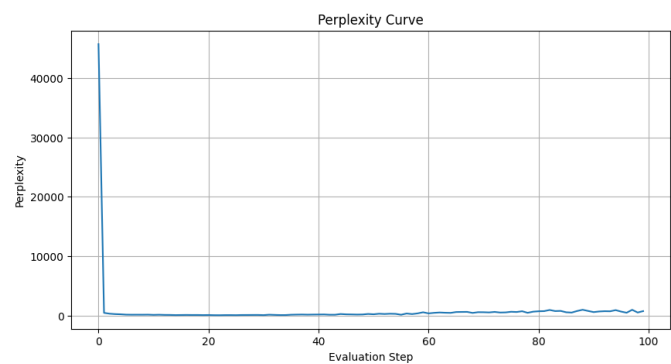


Fig. 4: Future scaling opportunities: Perplexity progression shows room for improvement through larger models and more data. Current 7.23 perplexity provides foundation for achieving 4–5 perplexity with scaled architecture.

- Persistent character consistency
- Complex plot development
- World lore integration

8) *Character-Specific Fine-Tuning*: Separate models could be trained for specific characters/archetypes:

- Wise wizard speech patterns
- Honorable knight dialogue
- Cunning rogue narratives
- Mystical elf perspectives
- Dwarf warrior descriptions

Overall, additional compute and data would primarily improve narrative consistency, long-range coherence, specialized world-building, and practical usability for interactive fantasy applications, while preserving domain specialization benefits.